Solution

(i) Differentiating $F(x)$ with respect to x at continuity points of $f(x)$, we get

$$f(x) = F'(x) = \begin{cases} 0, & x < 0 \\ 1, & 0 \le x < 1 \\ 0, & x \ge 1 \end{cases}$$

The pdf $f(x)$ is not continuous at $x = 0$, or at $x = 1$. We can define $f(0)$ and $f(1)$ in any manner. Choosing $f(0) = 1$, and $f(1) = 0$.

Therefore the probability density function $f(x)$ is

$$f(x) = \begin{cases} 1, & 0 \le x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(ii) $$P(0.2 \le X \le 0.7) = F(0.7) - F(0.2)$$
$$= 0.7 - 0.2 = 0.5$$

or

$$P(0.2 \le X \le 0.7) = \int_{0.2}^{0.7} f(x)\,dx = \int_{0.2}^{0.7} 1\,dx = 0.5 .$$

Remark

By definition, $P(X \le x) = F(x) = \int_{-\infty}^{x} f(u)\,du$. Probability $P(a < X < b)$ can be obtained by using either $F(x)$ or $f(x)$.

Note

We may also define the above probability density function as

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases} \text{ or } f(x) = \begin{cases} 1, & 0 < x \le 1 \\ 0, & \text{otherwise} \end{cases} \text{ or } f(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Example 11.14

The probability density function of random variable X is given by $f(x) = \begin{cases} k & 1 \le x \le 5 \\ 0 & \text{otherwise} \end{cases}$

Find (i) Distribution function (ii) $P(X < 3)$ (iii) $P(2 < X < 4)$ (iv) $P(3 \le X)$

Solution

Since $f(x)$ is a probability density function, $f(x) \ge 0$ and $\int_{-\infty}^{\infty} f(x)\,dx = 1$.

That is $$\int_{-\infty}^{1} 0\,dx + \int_{1}^{5} k\,dx + \int_{5}^{\infty} 0\,dx = 1 .$$

$$0 + k(x)_1^5 + 0 = 1 \Rightarrow 4k = 1 \Rightarrow k = \frac{1}{4} .$$

Therefore the probability density function is

$$f(x) = \begin{cases} \frac{1}{4}, & 1 \le x \le 5 \\ 0, & \text{otherwise} \end{cases}$$

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